Title:	Revision of the Bachelor of Science in Chemical Engineering and Revision of the Minor in Chemical Engineering
Sponsor:	*Department of Chemical Engineering*
Executive Summary:	There are 18 required courses in the College of Engineering that comprise the BS in Chemical Engineering. One of these courses is CHE 397: Senior Design II. The department proposes to increase the credit hours of this course from 3 to 3-4 hours to reflect the course load/level of effort required by students. The increase in one hour is being offset by changes to coursework in chemistry and calculus (which were approved by campus governance in April 2015 and will be in effect Fall 2016), and which reduce the hours of nonengineering and General Education coursework in the degree by one hour. As a result, the total hours required to earn the degree will remain at 128. Continuing students will take the course for 3 hours and students entering the degree in Fall 2016 will take the course for 4 hours. In addition, as the result of Department of Chemistry changes to the Honors General Chemistry sequence (CHEM 116 and 118) approved by the Senate in January 2016, analytical chemistry content will be incorporated into the lectures and labs of these two courses. As a result, chemical engineering students who complete CHEM 116 and 118 will not be required to take CHEM 222: Analytical Chemistry. However, they will be required to take an alternative chemistry course to meet nonengineering and General Education coursework requirements. Finally, the prerequisite courses for the Minor in Chemical Engineering are also being revised. The required courses that comprise the Minor will not change.
Description:	1. Effective Fall 2016, the credit hours of MATH 180: Calculus I and MATH 181: Calculus II will be reduced by one hour each (from 5 to 4 hours). In addition, the hours of CHEM 233: Organic Chemistry Laboratory will be increased by one hour (from 1 to 2 hours). As a result, the hours of nonengineering and General Education coursework in the degree will decrease by one hour (from 74 to 73 hours). *2.* Increase the credit hours for CHE 397: Senior Design II from 3 to 4. This results in a change to the Total Hours Required in College of Engineering from 48 to 49. As these changes in hours offset one another, the total hours required to earn the degree will remain at 128. 3. Effective Fall 2016, analytical chemistry content will be incorporated into the lectures and labs of the honors general chemistry sequence (CHEM 116 and 118). As a result, chemical engineering students who complete CHEM 116 and 118 to fulfill the general chemistry requirement will not be required to take CHEM 222: Analytical Chemistry. However, they will be required to take an alternative chemistry course to meet nonengineering and General

	Education coursework requirements. Specifically they may choose from among the following courses: • *CHEM 314 Inorganic Chemistry (4 hours), or* • *CHEM 452 Biochemistry (4 hours), or* • *CHEM 402 Chemical Information Systems (2 hrs) and CHEM 444 Physical Chemistry III (2 hrs)* Note: Honors general chemistry courses (CHEM 116/118) have long been allowed as substitutes for CHEM 112/114 (General College Chemistry I and II) – subsequently CHEM 122/123/124/125. This program revision also insures this practice is formally documented in the Undergraduate Catalog. 4. The Minor in Chemical Engineering requires prerequisite coursework primarily in chemistry, math and physics. The department proposes that two of these prerequisite courses be revised as follows: • CHEM 342: Physical Chemistry I be replaced with CHEM 342 or CHE 201: Introduction to Thermodynamics • CS 109: C/C++ Programming for Engineers with Mat Lab be replaced with CS 109 or CHE 205: Computational Methods in Chemical Engineering.
Justification:	1. Due to approved changes in the above MATH and CHEM courses from LAS, the undergraduate catalog needs to be revised to reflect these changes. 2. CHE 397 is the capstone chemical engineering design course; recent improvements in the course have increased the course load that justifies a 4-credit hour course. Also due to the change in CHE397 credit hours, the Total Hours Required in COE needs to be increased and the Non-Eng/GenEd requirements need to be decreased to reflect this change. 3. In order to maintain the appropriate number of hours of nonengineering and General Education coursework requires, Honors College students who complete CHEM 116 and 118 will need to complete an additional 4 hours of chemistry. CHEM 314, CHEM 452 and CHEM 402/CHEM 444 have been selected to satisfy the requirement because they will provide the Honors students with an additional understanding of chemistry by introducing biochemistry concepts, inorganic principles, or higher level physical chemistry theory and chemical information systems. In addition, these courses have been recommended by the Chemistry department as appropriate additional chemistry courses for the chemical engineering curriculum. 4. CHE 201 is a prerequisite for the 300-level CHE courses required in the minor, so it should be added to the prerequisite list as a selective. CHEM342 is sufficient for satisfying the CHE 201 prerequisite, particularly for LAS students to obtain a minor in CHE without the burden of an extra course. CHE 205 is a prerequisite for the 300-level CHE courses required in the minor, so it should be added to the prerequisite list as a selective. CS109 is sufficient for satisfying the CHE205 prerequisite and will allow students who have taken CS109 to obtain a minor in CHE without the burden of an extra course.

Catalog Statement:	*See below*
Minority Impact Statement:	*The changes will clarify the degree requirements for all students and will not adversely impact minority students.*
Budgetary and Staff Implications:	*N/A, as no new courses are being created.*
Library Resource Implications:	*N/A, as no new courses are being created.*
Space Implications:	*N/A*
Unit (e.g. department) approval date: *College (educational policy committee, faculty) approval dates:*	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering undergraduate committee approval; 11/20/15 Chemical engineering faculty approval: 1/28/16 Chemical Engineering department approval: 2/16/16 Education Policy Committee approval: 2/16/16
Contact Person:	Alan Zdunek zdunek@uic.edu
Proposed Effective Date/Term:	Fall, 2016

Current	Revised
BS in Chemical Engineering *Degree Requirements* To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the *College of Engineering* section for additional degree requirements and college academic policies. **BS in Chemical Engineering Degree Requirements Hours** Nonengineering and General Education Requirements 74	BS in Chemical Engineering SAME **BS in Chemical Engineering Degree Requirements Hours** Nonengineering and General Education Requirements **73**

Required in the College of Engineering	48
Technical Elective[a]	3
Electives outside the Major Rubric[a]	3
Total Hours—BS in Chemical Engineering	**128**

a *Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.*

Nonengineering and General Education Requirements

Courses	Hours
ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3
Exploring World Cultures course[a]	3
Understanding the Creative Arts course[a]	3
Understanding the Past course[a]	3
Understanding the Individual and Society course[a]	3
Understanding U.S. Society course[a]	3
MATH 180—Calculus I[b]	5
MATH 181—Calculus II[b]	5
MATH 210—Calculus III[b]	3
MATH 220—Introduction to Differential Equations	3
PHYS 141—General Physics I (Mechanics)[b]	4
PHYS 142—General Physics II (Electricity and Magnetism)[b]	4
CHEM 112—General College Chemistry I[b]	5
CHEM 114—General College Chemistry II[b]	5
CHEM 222—Analytical Chemistry	4
CHEM 232—Organic Chemistry I	4
CHEM 233—Organic Chemistry Laboratory I	1
CHEM 234—Organic Chemistry II	4
CHEM 342—Physical Chemistry I	3
CHEM 346—Physical Chemistry II	3
Total Hours—Nonengineering and General Education Requirements	**74**

a Students should consult the General Education section of the catalog for a list of approved courses in this category.
b This course is approved for the Analyzing the Natural World General Education category.

Required in the College of Engineering	**49**
Technical Elective[a]	3
Electives outside the Major Rubric[a]	3
Total Hours—BS in Chemical Engineering	**128**

a *Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.*

Nonengineering and General Education Requirements

ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3
Exploring World Cultures course[a]	3
Understanding the Creative Arts course[a]	3
Understanding the Past course[a]	3
Understanding the Individual and Society course[a]	3
Understanding U.S. Society course[a]	3
MATH 180—Calculus I[b]	**4**
MATH 181—Calculus II[b]	**4**
MATH 210—Calculus III[b]	3
MATH 220—Introduction to Differential Equations	3
PHYS 141—General Physics I (Mechanics)[b]	4
PHYS 142—General Physics II (Electricity and Magnetism)[b]	4
CHEM 122- General College Chemistry I[b] (Lecture)	**4**
CHEM 123- General College Chemistry I[b](Laboratory)	**1**
OR CHEM 116 – Honors and Major General and Analytical Chemistry I	**5**
CHEM 124—General College Chemistry I[b](Lecture)	**4**
CHEM 125—General College Chemistry II[b](Lab)	**1**
OR CHEM 118 – Honors and Major General and Analytical Chemistry II	**5**
CHEM 222—Analytical Chemistry[c]	4
CHEM 232—Organic Chemistry I	4
CHEM 233- Organic Chemistry Laboratory	**2**
CHEM 234—Organic Chemistry II	4
CHEM 342—Physical Chemistry I	3
CHEM 346—Physical Chemistry II	3
Total Hours- Nonengineering and General Education Requirements	**73**

a Students should consult the General Education section of the catalog for a list of approved courses in this category.
b This course is approved for the Analyzing the Natural World General Education category.
c Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222. Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402: Chemical Information Systems and CHEM 444: Physical Chemistry III.

Required in the College of Engineering	***Required in the College of Engineering***
	SAME
Courses — **Hours**	
ENGR 100—Orientation[a] — 0[a]	
CHE 201—Introduction to Thermodynamics — 3	
CHE 205—Computational Methods in Chem. Eng. — 3	
CHE 210—Material and Energy Balances — 4	
CHE 301—Chemical Engineering: Thermodynamics — 3	
CHE 311—Transport Phenomena I — 3	
CHE 312—Transport Phenomena II — 3	
CHE 313—Transport Phenomena III — 3	
CHE 321—Chemical Reaction Engineering — 3	
CHE 341—Chemical Process Control — 3	
CHE 381—Chemical Engineering Laboratory I — 2	
CHE 382—Chemical Engineering Laboratory II — 2	
CHE 396—Senior Design I — 4	
CHE 397—Senior Design II — 3	**CHE 397—Senior Design II — 4**
CHE 499—Professional Development Seminar — 0	**SAME**
ECE 210—Electrical Circuit Analysis — 3	
CME 260—Properties of Materials — 3	
CS 109—C/C++ Programming for Engineers with MatLab — 3	
Total Hours—Required in the College of Engineering — 48	**Total Hours—Required in the College of Engineering — 49**
a ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation.	SAME
Technical Elective	
Courses — **Hours**	SAME
One technical elective to be chosen from the following list of design-oriented courses:[a] — 3	
CHE 392—Undergraduate Research (3)[b]	
CHE 410—Transport Phenomena (3)	
CHE 413—Introduction to Flow in Porous Media (3)	
CHE 421—Combustion Engineering (3)	
CHE 422—Biochemical Engineering (3)	
CHE 423—Catalytic Reaction Engineering (3)	
CHE 425 Nanotechnology Drug Deliver (3)	
CHE 431—Numerical Methods in Chemical Engineering (3)	
CHE 433 Process Simulation with Aspen Plus (3)	
CHE 438—Computational Molecular Modeling (3)	
CHE 440—Non-Newtonian Fluids (3)	
CHE 441—Computer Applications in Chemical Engineering (3)	
CHE 445—Mathematical Methods in Chemical Engineering (3)	
CHE 450—Air Pollution Engineering (4)	
CHE 456—Fundamentals and Design of Microelectronics Processes 930	
CHE 494—Selected Topics in Chemical Engineering (3)	
Total Hours—Technical Elective — 3	

a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee.
b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering.

Courses	**Hours**
Electives outside the CHE rubric	3
Total Hours—Electives outside the Major Rubric	3

SAME

BS in Chemical Engineering, Bioengineering Concentration

Students in this concentration complete the following:

Required Courses [a]		
Technical Elective		
CHE 422	Biochemical Engineering	3
Electives		
Select two electives in nonmajor rubric category from the following:		5-7
BIOS 350	General Microbiology	
BIOS 351	Microbiology Laboratory	
CHEM 352	Introductory Biochemistry	
CHEM 452	Biochemistry I	
Total Hours		**8-10**

[a] *Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.*

SAME

Sample Course Schedule

Freshman Year

First Semester	**Hours**
MATH 180—Calculus I	5
CHEM 112—General College Chemistry I	5
ENGL 160—Academic Writing I: Writing for Academic and Public Contexts	3

Sample Course Schedule

Freshman Year

First Semester	**Hours**
MATH 180—Calculus I	**4**
CHEM 122—General College Chemistry I	**4**

CS 109—C/C++ Programming for Engineers with MatLab	3	**(Lecture)**	
		CHEM 123—General College Chemistry I (Lab)	**41**
		OR CHEM 116 – Honors and Major General and Analytical Chemistry I	5
ENGR 100—Orientation[a]	1[a]	ENGL 160—Academic Writing I: Writing for Academic and Public Contexts	3
		CS 109—C/C++ Programming for Engineers with MatLab	3
		ENGR 100—Orientation[a]	1[a]
Total Hours	**16**	**Total Hours**	**15**
Second Semester	**Hours**	**Second Semester**	**Hours**
MATH 181—Calculus II	5	**MATH 181—Calculus II**	**4**
PHYS 141—General Physics I (Mechanics)	4	PHYS 141—General Physics I (Mechanics)	4
ENGL 161—Academic Writing II: Writing for Inquiry and Research	3	ENGL 161—Academic Writing II: Writing for Inquiry and Research	3
CHEM 114—General College Chemistry II	5	**CHEM 124—General College Chemistry II (Lecture)**	**4**
		CHEM 125—General College Chemistry II (Lab)	**1**
		OR CHEM 118– Honors and Major General and Analytical Chemistry II	5
Total Hours	**17**	**Total Hours**	**16**
Sophomore Year			
First Semester	**Hours**	SAME	
MATH 210—Calculus III	3		
PHYS 142—General Physics II (Electricity and Magnetism)	4		
CHEM 232—Organic Chemistry I	4		
CHE 201—Introduction to Thermodynamics	3		
General Education Core course	3		
Total Hours	**17**		
Second Semester	**Hours**	**Second Semester**	**Hours**
MATH 220—Introduction to Differential Equations	3	MATH 220—Introduction to Differential Equations	3
CHEM 234—Organic Chemistry II	4	CHEM 234—Organic Chemistry II	4
CHEM 233—Organic Chemistry Lab I	1	**CHEM 233—Organic Chemistry Lab I**	**2**
CHE 205—Computational Methods in Chem. Eng.	3	CHE 205—Computational Methods in Chem. Eng.	3
CHE 210—Material and Energy Balances	4	CHE 210—Material and Energy Balances	4
CME 260— Properties of Materials	3	CME 260— Properties of Materials	3
Total Hours	**18**	**Total Hours**	**19**
Junior Year		SAME	
First Semester	**Hours**		
ECE 210—Electrical Circuit Analysis	3		
CHE 301—ChE Thermodynamics	3		
CHE 311—Transport Phenomena I	3		
CHEM 222—Analytical Chemistry	4		
General Education Core course	3		
Total Hours	**16**		

Second Semester	Hours
CHEM 342—Physical Chemistry I	3
CHE 312—Transport Phenomena II	3
CHE 313— Transport Phenomena III	3
CHE 321—Chemical Reaction Engineering	3
General Education Core course	3
Total Hours	**15**

Senior Year

First Semester	Hours
CHE 381—Chemical Engineering Laboratory I	2
CHE 396—Senior Design I	4
CHEM 346—Physical Chemistry II	3
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 456, 494, or 392 (departmental approval is required for CHE 392)	3
General Education Core Course	3
Total Hours	**15**

Second Semester	Hours
CHE 382—Chemical Engineering Laboratory II	2
CHE 341—Chemical Process Control	3
CHE 397—Senior Design II	3
CHE 499—Professional Development Seminar	0
Elective outside the major rubric	3
General Education Core course	3
Total Hours	**14**

Minor in Chemical Engineering

For the minor, 16–18 semester hours are required, excluding prerequisite courses. Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete the following:

Prerequisite Courses		
CHEM 112 or CHEM 116	General College Chemistry I Honors General Chemistry I	5
CHEM 342	Physical Chemistry I	3
CS 109	C/C ++ Programming for Engineers with MatLab	3
MATH 180	Calculus I	5
MATH 181	Calculus II	5
MATH 210	Calculus III	3
MATH 220	Introduction to Differential Equations	3

SAME

SAME

Second Semester	Hours
CHE 382—Chemical Engineering Laboratory II	2
CHE 341—Chemical Process Control	3
CHE 397—Senior Design II	**4**
CHE 499—Professional Development Seminar	0
Elective outside the major rubric	3
General Education Core course	3
Total Hours	**15**

Minor in Chemical Engineering

For the minor, 16–18 semester hours are required, excluding prerequisite courses. Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete the following:

Prerequisite Courses		
CHEM 122	General College Chemistry I (Lecture)	4
CHEM 123	General College Chemistry I (Laboratory)	1
Or CHEM 116	Honors and Majors General and Analytical Chemistry I	5
CHEM 342 or CHE 301	Physical Chemistry I or Introduction to Thermodynamics	3 3

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PHYS 141	General Physics I (Mechanics)	4
PHYS 142	General Physics II (Electricity and Magnetism)	4
Total Hours		**35**

Required Courses

CHE 210	Material and Energy Balances	4
CHE 301	Chemical Engineering Thermodynamics	3
CHE 321	Chemical Reaction Engineering	3
CHE 311 or ME 211	Transport Phenomena I Fluid Mechanics I	3-4
Select one of the following:		3-4
CHE 312	Transport Phenomena II	
ME 321	Heat Transfer	
CHE 313	Transport Phenomena III	
Total Hours		**16-18**

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CS 109 or CHE 205	C/C ++ Programming for Engineers with MatLab or Computational Methods in Chemical Engineering	3 3
MATH 180	Calculus I	4
MATH 181	Calculus II	4

SAME

Total Hours **33**

SAME

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